

Q1

With MAP_PRIVATE

```
[asjadraza@Asjads-MacBook-Air-2 OS % gcc Q1.c -o Q1 && ./Q1
Value of var in parent: 10
asjadraza@Asjads-MacBook-Air-2 OS % █
```

With MAP_SHARED

```
[asjadraza@Asjads-MacBook-Air-2 OS % gcc Q1.c -o Q1 && ./Q1
Value of var in parent: 99
asjadraza@Asjads-MacBook-Air-2 OS % █
```

Previously the map was private, so the changes were reflected locally, upto the child fork only. But when it got shared, the changes reflect everywhere, in the parent too.

Q2

```
[asjadraza@Asjads-MacBook-Air-2 OS % gcc Q2.c -o Q2 && ./Q2
Child 1 incremented counter to: 1
Child 1 incremented counter to: 2
Child 1 incremented counter to: 3
Child 1 incremented counter to: 4
Child 1 incremented counter to: 5
Child 2 incremented counter to: 6
Child 2 incremented counter to: 7
Child 2 incremented counter to: 8
Child 2 incremented counter to: 9
Child 2 incremented counter to: 10
Value of counter in parent: 10
asjadraza@Asjads-MacBook-Air-2 OS % █
```

Q3

```
[asjadraza@Asjads-MacBook-Air-2 OS % gcc writer.c -o writer && ./writer
Writer completed.
[asjadraza@Asjads-MacBook-Air-2 OS % gcc reader.c -o reader && ./reader
Number: 14
Text: Hellllloooooooolooloolooloolooloo
asjadraza@Asjads-MacBook-Air-2 OS % █
```

Q4

```
asjadraza@Asjads-MacBook-Air-2 OS % gcc producer.c -o producer
asjadraza@Asjads-MacBook-Air-2 OS % ./producer 10
Producer: counter = 1
Producer: counter = 2
Producer: counter = 3
Producer: counter = 4
Producer: counter = 5
Producer: counter = 6
Producer: counter = 7
Producer: counter = 8
Producer: counter = 9
Producer: counter = 10
Producer finished.
asjadraza@Asjads-MacBook-Air-2 OS %
```

```
asjadraza@Asjads-MacBook-Air-2 OS % gcc consumer.c -o consumer
asjadraza@Asjads-MacBook-Air-2 OS % ./consumer
Consumer: counter = 1
Consumer: counter = 2
Consumer: counter = 3
Consumer: counter = 4
Consumer: counter = 5
Consumer: counter = 6
Consumer: counter = 7
Consumer: counter = 8
Consumer: counter = 9
Consumer: counter = 10
Final counter value: 10
asjadraza@Asjads-MacBook-Air-2 OS %
```